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ORIGINAL ARTICLE

Laboratory science

A global comparative field study to evaluate the factor VIII activity of efanesoctocog alfa by one-stage clotting and chromogenic substrate assays at clinical haemostasis laboratories

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Abstract

Introduction: Structural and chemical modifications of factor VIII (FVIII) products may influence their behaviour in FVIII activity assays. Hence, it is important to assess the performance of FVIII products in these assays. Efanesoctocog alfa is a new class of FVIII replacement therapy designed to provide both high sustained factor activity levels and prolonged plasma half-life.

Aim: Evaluate the accuracy of measuring efanesoctocog alfa FVIII activity in one-stage clotting assays (OSAs) and chromogenic substrate assays (CSAs).

Methods: Human plasma with no detectable FVIII activity was spiked with efanesoctocog alfa or a full-length recombinant FVIII product comparator, octocog alfa, at nominal concentrations of 0.80 IU/mL, 0.20 IU/mL, or 0.05 IU/mL, based on labelled potency. Clinical haemostasis laboratories (N = 35) tested blinded samples using in-house assays. Data from 51 OSAs (14 activated partial thromboplastin time [aPTT] reagents) and 42 CSAs (eight kits) were analyzed.

Results: Efanesoctocog alfa activity was reliably (±25% of nominal activity) measured across all concentrations using OSAs with Actin FSL and multiple other aPTT reagents. Under- and overestimation of FVIII activity occurred with some reagents. No specific trend was observed for any class of aPTT activators. A two- to three-fold overestimation was consistently observed using CSAs and the OSA with Actin FS as the aPTT reagent across evaluated concentrations.

Conclusion: Under- or overestimation occurred with some specific OSAs and most CSAs, which has been previously observed with other modified FVIII replacement products. Efanesoctocog alfa FVIII activity was measured with acceptable accuracy and reliability using several OSA methods and commercial plasma standards.

KEYWORDS
aPTT, assay, chromogenic, coagulation, FVIII replacement therapy, Haemophilia A, von Willebrand factor

1 | INTRODUCTION

Prophylaxis is the standard of care for patients with severe haemophilia A, with a primary aim to provide sufficient factor VIII (FVIII) activity levels to reduce the risk of spontaneous bleeds and prolonged bleeds after trauma.[1] In clinical practice, measuring plasma FVIII activity is important for diagnosis and severity assessment, monitoring of FVIII replacement therapy, and evaluating the need for FVIII supplementation during surgery and management of severe bleeds.[1,2]

Measurement of FVIII activity levels is typically performed in clinical haemostasis laboratories using one of two types of assays: one-stage clotting assay (OSA) and chromogenic substrate assay (CSA).[2] Both types are calibrated against plasma FVIII standards, but the methodological differences result in variability and discordance in their performance.[2] Additionally, discrepancies in results within each assay type have been widely documented and may be attributed to the use of a variety of reagents, analyzers, assay manufacturers, and methods by clinical haemostasis laboratories around the world.[3–8]

Further variability in assay performance, seen with different FVIII replacement products, may be due to specific modifications to the FVIII molecule itself (such as recombinant production, B-domain deletion, dual/single chain, PEGylation).[5–7,9–14] Different results between one-stage clotting assays (OSAs) and chromogenic substrate assays (CSAs), and their reagents, have been reported for many modified FVIII products, including antihaemophilic FVIII (recombinant) (Kovaltry®),[6] damoctocog alfa pegol (Jivi®),[10,13,15] turoctocog alfa pegol (Esperoct®),[12,16–19] moroctocog alfa (ReFacto AF®),[3,5,20,21] and lonoctocog alfa (Afstyla®).[11,13,22] Although full-length FVIII products have typically shown ∼20% higher FVIII activity with the CSA than with the OSA,[6] the difference may be as high as ∼50% for the B-domain deleted Refacto AF® and Afstyla®.[3,11] As such variabilities can impact dosing in clinical management,[23] it is important to assess how each new FVIII replacement product will be measured by different assays in the field. Recommendations have been published to assist laboratories and haemophilia treaters in the successful management of such variabilities.[1]

Efanesoctocog alfa (ALTUVIIIO™, formerly BIVV001) is a new class of high sustained FVIII replacement therapy that circulates in blood independently of endogenous von Willebrand factor (VWF). It is composed of a single B-domain deleted FVIII protein, the FVIII-binding D′D3 domain of VWF, IgG1-Fc domain and two XTEN polypeptides (Figure 1).[24] Various in vitro and in vivo efficacy experiments have confirmed that efanesoctocog alfa potency is best assigned by the OSA with Actin FSL as reagent[24,25]; when measured by a panel of CSAs, efanesoctocog alfa has ∼2.5-fold higher molar specific activity than that measured by an OSA using Actin FSL.[24] Therefore, the OSA with Actin FSL was the primary method used in efanesoctocog alfa clinical trials.[26–28] When assessed by OSA with Actin FSL, the potency of efanesoctocog alfa in promoting functional clot formation is indistinguishable from that of recombinant FVIII.[25]

In a Phase 3 clinical trial (XTEND-1), once-weekly prophylaxis intravenous dose with 50 IU/kg efanesoctocog alfa provided high sustained mean FVIII activity levels in the normal to near-normal range (>40 IU/dL) for the majority of the week and superior bleed protection in patients with severe haemophilia A versus prior FVIII prophylactic replacement therapy.[28] Here, we present the results from a comparative assay field study for which the objective was to evaluate the performance of OSA and CSA assays in the measurement of FVIII activity of efanesoctocog alfa in plasma samples using a variety of OSA reagents, CSA kits and analyzers in clinical haemostasis laboratories in multiple countries.

2 | MATERIALS AND METHODS

2.1 | Design

Participating clinical haemostasis laboratories ($N = 35$) from 13 countries (Supplementary Figure S1) received field study samples and the study was conducted between May 2021 and April 2022. All clinical laboratories were instructed to test all plasma samples using their routine FVIII activity procedures and calibrators (commercially available plasma or pooled normal plasma calibrated against the World Health Organization [WHO] international FVIII reference plasma). Laboratories were blinded to the sample contents in the kits. To assess intra-laboratory variability, each sample was required to be tested in triplicate over three separate days (i.e., nine reports per sample per assay in total).

2.2 | Field study sample kits

Field study sample kits were manufactured and distributed to participating laboratories by Precision Biologics, Inc. FVIII inhibitor free congenital FVIII-deficient pooled plasma (purchased from HRF Inc., Raleigh, NC) with no detectable FVIII activity (<0.005 IU/mL), was spiked with a single lot of either efanesoctocog alfa or the marketed full-length recombinant FVIII product comparator, octocog alfa, at nominal levels of 0.80, 0.20, or 0.05 IU/mL, based on labelled potency. For efanesoctocog alfa, labelled potency was assigned based on the OSA using Actin FSL as the reagent on automated blood coagulation analyzer Sysmex CA-1500 with WHO 8th International Standard for FVIII Concentrate (NIBSC code 07/350).[24] Each kit included 18 tubes

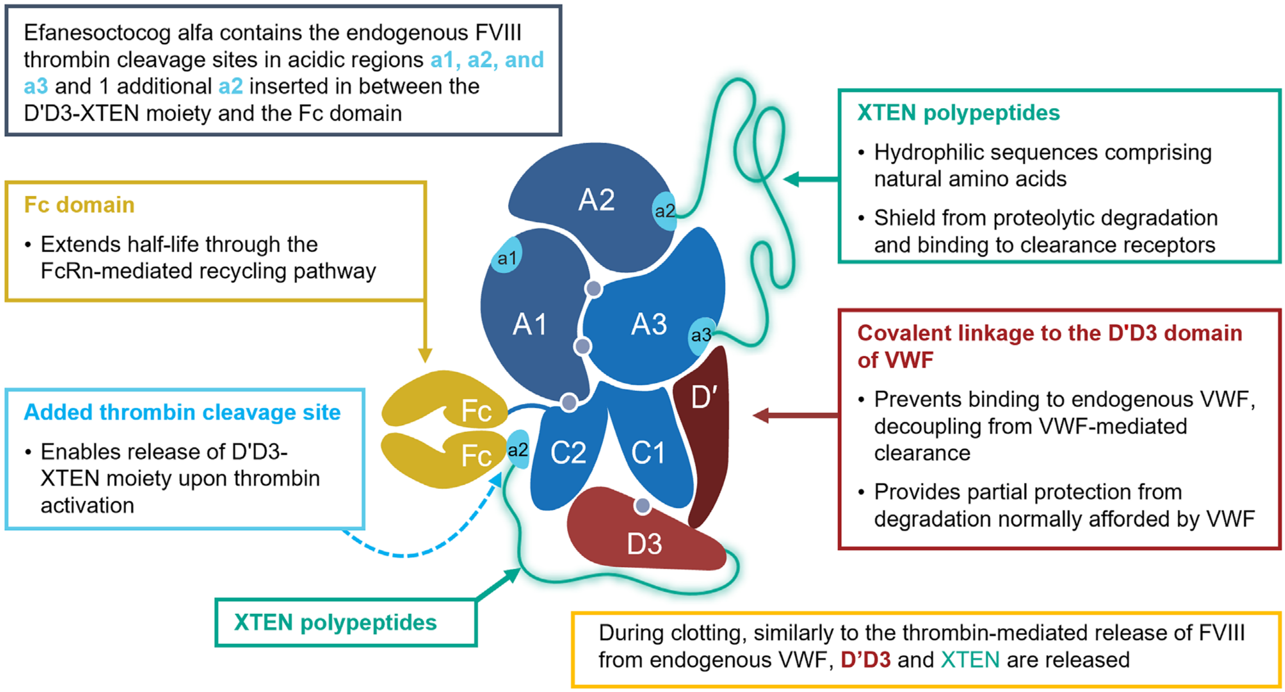

FIGURE 1 Structure and molecular activation of efanesoctocog alfa. Grey circles indicate divalent cations. a1, a2, a3, acidic region 1, 2, and 3; Fc, fragment crystallizable region of immunoglobulin G1; FcRn, neonatal Fc receptor; FVIII, factor VIII; rFVIIIFc, recombinant clotting factor VIII Fc-fusion protein; VWF, von Willebrand factor. Additional a2 site was introduced between D'D3XTEN-Fc to allow release of D'D3-XTEN upon efanesoctocog alfa activation with thrombin. This a2 site has the same amino acid sequence as the a2 site in FVIII. From BIVV001 Fusion Protein as Factor VIII Replacement Therapy for Hemophilia A. Konkle BA, et al. N Eng J Med. 2020;383:1018–1027. Copyright © (2022) Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

containing three aliquots each of the three activity levels for both octocog alfa–spiked plasma (comparator) and efanesoctocog alfa–spiked test plasma. Samples were distributed as frozen aliquots to resemble patient plasma samples stored in laboratories for routine testing and to mitigate pre-analytic variability associated with sample reconstitution.

The effect of freeze-thaw cycles on the FVIII activity of efanesoctocog alfa and octocog alfa has been evaluated. In the control experimental conditions, OSA (Siemens Actin FSL) and CSA (Hyphen Biomed Biophen FVIII) activity of efanesoctocog alfa and octocog alfa (~0.8, 0.2, 0.05 IU/mL in 1 mL congenital FVIII-deficient plasma) appeared to be stable up to 3 freeze-thaw cycles (thawing in a 37°C water bath for 4 min, sample placed on ice after thawing and frozen in liquid nitrogen below −70°C within 30 min of thawing). Recovered activities for all samples even after further freeze-thaw cycles remained within ±20% of initial measurements. However, to avoid variability in the freeze-thaw process or mishandling of samples, we do not recommend more than one freeze-thaw cycle.

2.3 | FVIII activity measurement with OSA and CSA

Each laboratory reported detailed information on their assay components and procedures including the type and source of OSA reagents or CSA kits, analyzer employed, substrate plasma, number of dilutions performed in each assay, and source of calibrators. For both OSAs and CSAs, most analyzers and reagents used were manufacturer-recommended: in total six different classes of analyzers were used.

2.3.1 | OSA

Data from 51 OSAs were available for analysis. Eleven laboratories tested more than one OSA reagent. Fourteen different commercially available activated partial thromboplastin time (aPTT) reagents were used, the most common of which were SynthASil (n = 15), Actin FS (n = 10), and Actin FSL (n = 7). Four types of aPTT activators were used, the most common of which were silica (n = 25) and ellagic acid (n = 21) (Figure S2).

2.3.2 | CSA

Data from 42 CSAs were available for analysis. All but one of the 35 laboratories performed at least one CSA. Altogether, eight different CSA kits were used, which utilized human- (n = 9; Hyphen, also known as Hyphen Biomed kit, was the only human kit), bovine- (n = 13), or hybrid-based components/catalysts (n = 20) (Figure S2).

2.4 | Data analysis

Raw data and the final FVIII activity results for each sample provided by the laboratories were reviewed and cross-checked for accuracy and consistency before data analysis was conducted.

Results were analysed for intra-assay (intra-day and inter-day) and inter-assay variation. Inter-assay variability was evaluated with respect to the type of activator/phospholipids in the commercial aPTT reagents, number of sample dilutions, calibrators, and analyzer used.

FVIII recovery was calculated by dividing the mean activities obtained in an assay with the assigned target nominal value, represented as percentage of nominal.

Statistical evaluation was performed by analysis of variance (ANOVA) to compare FVIII activity when using different assay reagents, analyzers, or procedures.

2.5 | Quality assurance

The majority of laboratories had a Clinical Laboratory Improvement Amendments (CLIA)[29] or equivalent certification and participated in external proficiency programs. By utilizing daily quality control testing, laboratories had confidence in the performance of their existing FVIII assays. Data reporting and analyses were conducted in accordance with good documentation practices at an ISO 13485 certified company (Precision BioLogic).[30]

3 | RESULTS

3.1 | OSA results

The OSA results (n = 51) from various laboratories are presented in Figure 2. Overall, for all efanesoctocog alfa samples and all OSA reagents, the combined mean measured FVIII activity values for the expected (nominal) values of 0.80, 0.20, and 0.05 IU/mL were 0.925 IU/mL, 0.266 IU/mL, and 0.080 IU/mL, respectively (Table 1). The mean estimated recovery of nominal activity was 116%, 133%, and 160%, respectively.

Reagent-correlated variability was within the acceptable range of 100% ± 25% for octocog alfa across most OSA reagents. For octocog alfa, the combined mean measured FVIII activity values for the expected (nominal) values of 0.80, 0.20, and 0.05 IU/mL were 0.675, 0.182, and 0.052 IU/mL, respectively (Table 1). The mean estimated recovery of nominal activity was 84%, 91%, and 104%, respectively.

3.1.1 | Actin FSL

When tested using OSA with Actin FSL (n = 7), the reagent used for potency assignment, the efanesoctocog alfa FVIII activity was reliably (± 25% of nominal activity) measured across all activity levels (Table S1, Figure 2), with a linear dose-dependent recovery.

3.1.2 | SynthASil

When measured using SynthASil (n = 15), mean efanesoctocog alfa FVIII activity was underestimated by 30%–40% for the 0.80 and 0.20 IU/mL samples (Figure 2).

3.1.3 | Actin FS

When tested using OSA with Actin FS (n = 10), mean efanesoctocog alfa FVIII activity was overestimated by 225%–285% across all activity levels (Table S1, Figure 2).

If Actin FS values were excluded from the OSA results, the combined mean of measured FVIII activity of the remaining results (n = 41) for the expected (nominal) values of 0.80, 0.20, and 0.05 IU/mL were reduced to 0.694 IU/mL, 0.207 IU/mL, and 0.065 IU/mL, respectively (Table 1). This resulted in the mean estimated recovery of 87%, 104%, and 130%, respectively, which were almost all within the expected range with respect to nominal activity.

3.1.4 | Other reagents

Reliable results for the high efanesoctocog alfa level were observed at laboratories using the ellagic acid-based (SynthAFax, Revohem SLA, Thrombocheck SLA), silica-based (Pathromtin SL, PTT-A, aPTT-HS), and kaolin-based (CK-Prest) reagents, with some over-estimation observed at lower levels by these reagents (Figure 2). Other rarely used reagents by laboratories in this study (APTT SP [n = 2], Actin [n = 1], Cephascreen [n = 1], and Cephen [n = 1]) showed some over- or underestimation.

No specific trend was observed for any class of aPTT activators or analyzers.

3.1.5 | Laboratory variability

The variability for intra- and inter-day testing in each laboratory for both efanesoctocog alfa and octocog alfa was <10% coefficient of variation (CV) (Table S2). Inter-assay variability was higher for efanesoctocog alfa than octocog alfa (Table 1). No specific trend was observed when affiliated reagent-analyzer combination versus non-affiliated was used (Figure S3).

3.2 | CSA results

The CSA data (n = 42) from various laboratories are presented in Figure 3. Overall, across all CSAs, the combined mean of measured

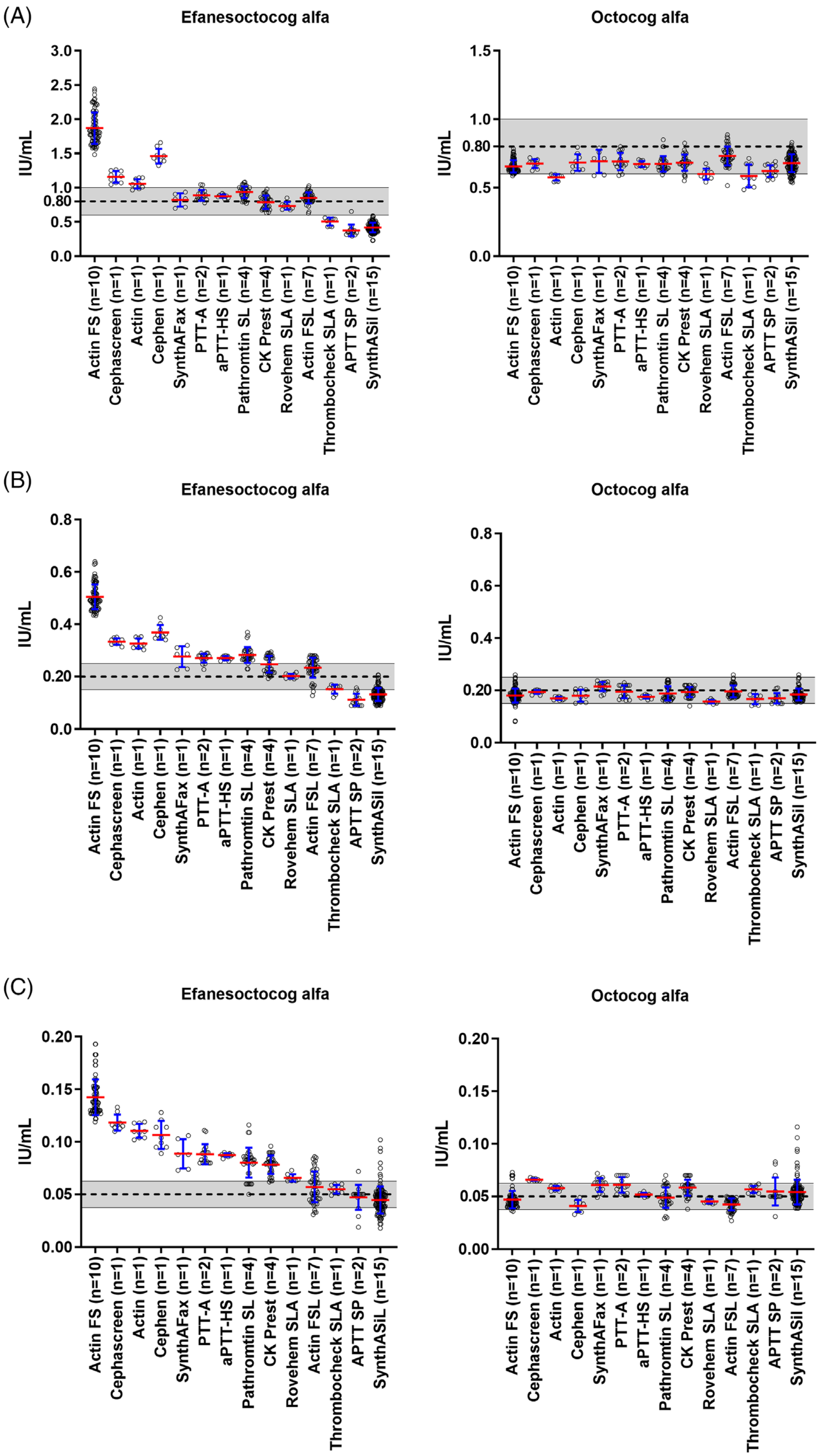

FIGURE 2 OSA FVIII activity at (A) 0.80, (B) 0.20, and (C) 0.05 IU/mL (51 assays). aPTT, activated partial thromboplastin time; FVIII, factor VIII; OSA, one-stage clotting assay; SD, standard deviation. Gray shaded areas show ± 25% of nominal activity; mean and ± 1 SD presented by red and blue lines.

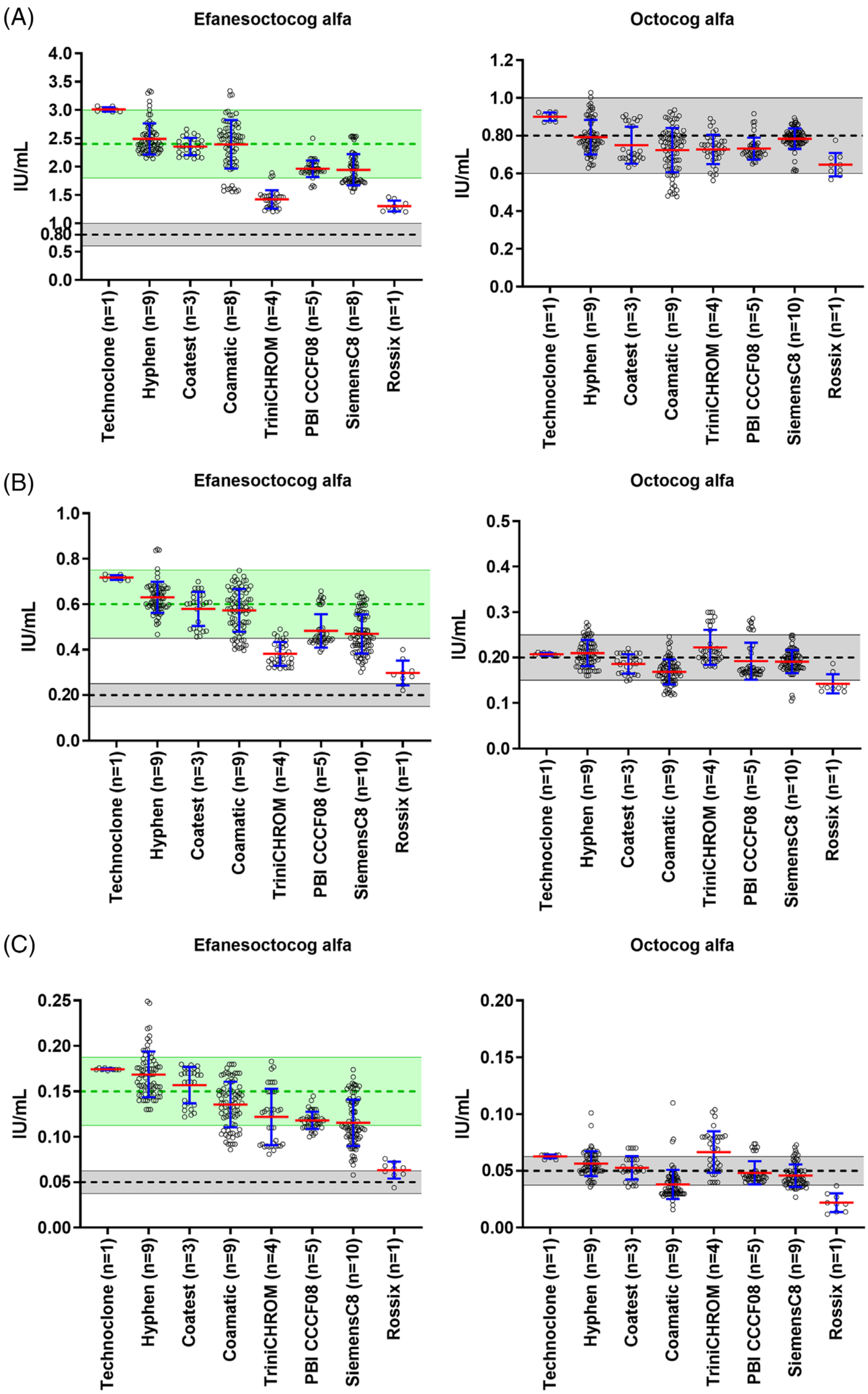

FIGURE 3 CSA FVIII activity at (A) 0.80, (B) 0.20, and (C) 0.05 IU/mL (42 assays). CSA, chromogenic substrate assay; FVIII, factor VIII; SD, standard deviation. Grey and green shaded areas show ± 25% of 1× and 3× nominal activity, respectively; red and blue lines represent mean and ± 1 SD, respectively. A few sites reported CSA results as out of their measurement range for the 0.80 IU/mL sample.

TABLE 1 Arithmetic mean and overall variations (mean CV%) of OSA and CSA FVIII assays in measurement of efanesoctocog alfa samples and octocog alfa controls.

					Excluding Actin FS assays ($N = 41$)	
Assay	Product	Nominal sample concentration (IU/mL)	Mean (IU/mL)	Inter-assay coefficient of variation (%)	Mean (IU/mL)	Inter-assay coefficient of variation (%)
OSA FVIII ($N = 51$)	Octocog alfa	0.05	0.052	19.2	0.053	19.3
		0.20	0.182	12.1	0.185	11.8
		0.80	0.675	8.0	0.680	8.4
	Efanesoctocog alfa	0.05	0.080	47.8	0.065	36.2
		0.20	0.266	52.3	0.207	36.8
		0.80	0.925	58.5	0.694	39.8
CSA FVIII ($N = 42$)[a]	Octocog alfa	0.05	0.048	30.0	N/A	N/A
		0.20	0.193	16.7	N/A	N/A
		0.80	0.759	11.0	N/A	N/A
	Efanesoctocog alfa	0.05	0.136	24.3	N/A	N/A
		0.20	0.530	22.1	N/A	N/A
		0.80	2.158	21.4	N/A	N/A

Abbreviations: CSA, chromogenic substrate assay; CV, coefficient of variation; FVIII, factor VIII; N/A, not applicable; OSA, one-stage clotting assay.
[a]Three sites reported CSA results as out of their measurement range (related to the 0.80 IU/mL efanesoctocog alfa sample).

efanesoctocog alfa FVIII activity for the expected (nominal) values of 0.80, 0.20, and 0.05 IU/mL was 2.158 IU/mL, 0.530 IU/mL, and 0.136 IU/mL, respectively (Table 1). The mean estimated recovery with respect to nominal activity was 270%, 265%, and 271%, respectively.

For octocog alfa, the mean FVIII recovery by all CSAs was within 100% ± 25% range for 0.8 IU/mL and 0.2 IU/mL, while recovery was within 100% ± 30% for 0.05 IU/mL.

3.2.1 | TriniCHROM and Rossix

For TriniCHROM and Rossix CSAs, efanesoctocog alfa FVIII activity recovery was slightly lower than with other CSAs. For TriniCHROM ($n = 4$), estimated recovery was 180%–190% at 0.80 and 0.20 IU/mL, and 250% at 0.05 IU/mL. For Rossix ($n = 1$), estimated recovery was 130%–160% at all three activity levels.

Excluding Rossix, mean estimated FVIII recovery for the most commonly used CSA kits across all three activity levels was 271% of the nominal potency value.

3.2.2 | Laboratory variability

Intra- and inter-day variability within a laboratory for the CSA was <10% CV for both efanesoctocog alfa and octocog alfa (Table S2).

Inter-assay variability across different CSA kits ranged from 11% to 30% CV for octocog alfa and from 21% to 24% CV for efanesoctocog alfa. No specific trend was observed when reagent-analyzer combination was used (Figure S4).

Overall, OSA activity with Actin FSL recovered within the expected range, while the mean ratio of CSA FVIII activity recovery with respect to Actin FSL assigned nominal activity was 2.7 (range 2–3) (Figure 4). A similar difference was observed between Actin FS and Actin FSL.

3.3 | Excluded data sets

Two data sets were excluded from the analyses as outliers: data from one study site using Siemens OSA Actin FSL/BCSXP was excluded due to deficiency in procedure and improper dilution used for testing the samples, and data from another study site using IL SynthAFax/Stago Evolution was excluded due to an extremely high value compared with testing on IL TOP analyzer, which were within ±25% of nominal values. For one lab, which tested more than one reagent, results were included from one assay only as results from additional assays were not received before the study closure.

4 | DISCUSSION

This global comparative field study analysed the performance of 51 OSAs and 42 CSAs in measuring efanesoctocog alfa FVIII activity, in 35 clinical haemostasis laboratories. Reliable results for efanesoctocog alfa were observed with several OSAs (±25% of nominal activity), despite the use of a wide variety of aPTT reagents, congenital FVIII-deficient substrate plasma, sample dilutions, calibrators, and operating procedures. As expected, based on a two- to three-fold difference in specific activity between OSA and CSA,[24] an accordingly important

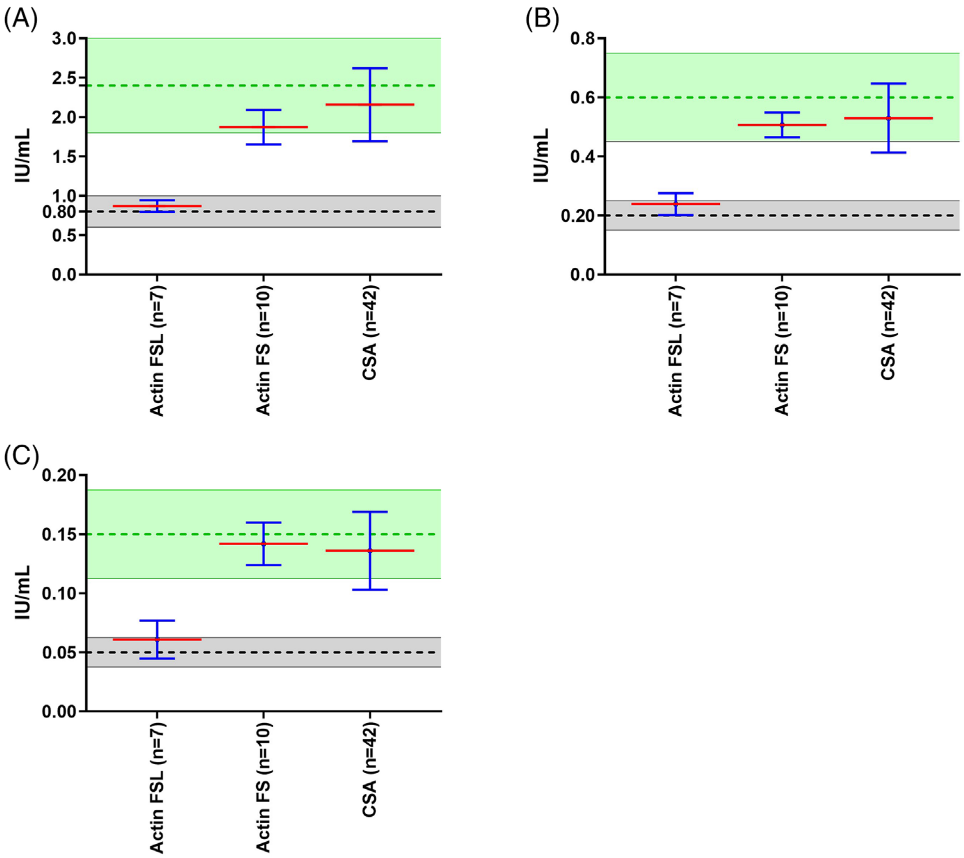

FIGURE 4 Efanesoctocog alfa recovery at (A) 0.80, (B) 0.20, and (C) 0.05 IU/mL: OSA (based on Actin FS, Actin FSL) and CSA. CSA, chromogenic substrate assay; OSA, one-stage clotting assay; SD, standard deviation. Gray and green shaded areas show ± 25% of 1× and 3× nominal activity. Red and blue lines represent mean and ± 1 SD, respectively. The average recovery across the three concentrations for OSA with Actin FS was 254.3%, and for CSA was 268.5%.

overestimation was observed with most CSA kits across all activity levels.

Globally, the OSA remains the most commonly used method for measuring FVIII activity in clinical laboratories.[1] Reliable results for efanesoctocog alfa FVIII activity were observed with Actin FSL, SynthAFax, Thrombocheck SLA, Pathromtin SL, PTT-A, aPTT-HS, and CK-Prest OSAs, although some overestimation was seen at lower activity levels. There was no specific correlation in over- or underestimation with a particular class of aPTT activators (ie, ellagic acid, kaolin, silica), and no specific trend was observed when reagent-analyzer combination was used. Actin FSL was the optimum OSA reagent, with the least variability across the range. In contrast, a clinically important overestimation was observed with Actin FS across all three activity levels and some underestimation with SynthASil; similar findings have been reported for other FVIII replacement therapies.[10,31] The observed differences in OSA and CSA performance may be at least partly attributed to variations in assay reagents, incubation times, and thrombin concentration impacting the activation kinetics of efanesoctocog alfa.[24]

A previous study reported that Actin FS has higher sensitivity to factor deficiencies compared with Actin FSL.[32] Differences in phospholipid content may account for the relatively high sensitivity of Actin FS for efanesoctocog alfa FVIII activity. While Actin FS and Actin FSL both use ellagic acid (0.1 mM) as activator, Actin FS contains purified phosphatides from soy and Actin FSL contains soy and the more traditional rabbit brain as the source of phospholipids.[33] In addition, the composition of fatty acids of Actin FS differs from that of other reagents, with a higher ratio of unsaturated to saturated fatty acids and absence of phosphatidyl serine.[33] When used to measure efanesoctocog alfa FVIII activity, an OSA with Actin FS behaves similarly to CSA, which has been observed previously with other FVIII therapies.[34]

The performance of CSAs is generally independent of FVIII activation kinetics, due to the presence of excess thrombin and longer incubation times than with OSAs, and they are typically associated with less inter-laboratory variability.[6] In this study, most CSAs showed a similar level of overestimation of efanesoctocog alfa. However, TriniCHROM and Rossix CSAs had less over-estimation than other CSA kits, perhaps due to variations in the incubation times/thrombin concentrations in these kits. The difference in performance between CSAs and OSAs in this study may be partially explained by the fact that efanesoctocog alfa is a single-chain FVIII and also requires additional thrombin cleavage events to become fully activated (an additional acidic region site [a2] was introduced between D'D3XTEN-Fc to allow release of D'D3-XTEN polypeptide).[9,24]

At this time, performance of efanesoctocog alfa FVIII activity with OSAs using APTT SP, Actin, Cephen, or Cephascreen cannot be reliably confirmed, due to low numbers of laboratories that used these reagents.

In summary, under- or overestimation is expected to occur across some OSA assays, and inter-assay variability has been observed for several other modified FVIII replacement therapies. The ALTUVIIIO package insert recommends using a validated OSA to measure efanesoctocog alfa FVIII activity and, if an OSA with a specific ellagic acid based aPTT reagent (i.e., Actin FS) or a CSA is used, it also recommends that the result is divided by 2.5 to approximate FVIII activity level.[35] This correction factor does not apply if an OSA is performed with other reagents. Routine monitoring in haemophilia A is becoming less prevalent, particularly with new non-factor therapies.[36] Although it is not anticipated that routine monitoring of FVIII activity during efanesoctocog alfa prophylaxis will be needed in clinical practice due to its VWF-independent mechanism of half-life extension and low inter-subject half-life variability,[37] FVIII activity measurement will be necessary in certain clinical circumstances. Thus, as with all new FVIII replacement products, it is important to understand the real-world assay variability in measuring efanesoctocog alfa FVIII activity and for individual laboratories to determine the best in-house approach for FVIII testing.[38] Our study showed that efanesoctocog alfa FVIII activity is reliably measured by the commonly available OSA using multiple aPTT reagents, but when using Actin FS–based OSA or the CSA, a two- to three-fold overestimation should be expected.

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CONFLICT OF INTEREST STATEMENT

AW, ML, and KK are employees of Sanofi. ESC is an employee of Sanofi and holds shares in the company. ASK is an employee of Precision BioLogic Inc. SWP has served as a consultant to Apcintex, ASC Therapeutics, Bayer, BioMarin, CSL Behring, GeneVentiv, HEMA Biologics, Freeline, LFB, Novo Nordisk, Pfizer, Regeneron/Intellia, Roche/Genentech, Sanofi, Takeda, Spark Therapeutics, and uniQure. BAK has served as a paid consultant to Sanofi, BioMarin, Pfizer, and NovoNordisk and has received research funding from Sanofi, CSL Behring/Uniqure, Pfizer, Spark, and Takeda. SK has acted as a paid consultant to Sanofi. CN is a consultant for Sobi and has received education and speaking fees for Novo Nordisk. ES and LAF are employees at Sobi.

DATA AVAILABILITY STATEMENT

Qualified researchers may request access to data and related study documents. Further details on Sanofi's data sharing criteria, eligible studies, and process for requesting access can be found at https://www.vivli.org.

ETHICS STATEMENT

This manuscript reports outcomes from nonclinical in vitro experiments and did not utilize animals or human subjects.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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